Implementation of Peer-to-Peer Energy Trading via Solana Smart Contracts Permissioned Environment

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Thailand's state utilities EGAT, MEA, and PEA have advanced a National Energy Trading Platform (NETP) since 2017, envisaging a blockchain-based settlement layer with a dedicated digital currency for Peer-to-Peer (P2P) trading among rooftop-solar prosumers. The platform remains in development, and the regulated buy-back scheme still gives prosumers no direct mechanism to exchange surplus generation within the local distribution grid. This paper presents a smart-contract-based on-chain settlement layer for P2P energy trading, realized as six modular Anchor programs deployed on a permissioned Solana runtime that uses the Sealevel parallel-execution engine. By routing every high-frequency write to a per-entity program-derived account (PDA), the system lets transactions with disjoint write sets execute in parallel, with transaction ordering provided by Solana's Proof-of-History (PoH). The market-clearing framework couples a continuous double auction (CDA) for real-time trades with a uniform-price auction that clears aggregated prosumer bids at a single settlement price over fifteen-minute epochs. We simulate consumption, generation, and power flow for 80 meters over 30 days at fifteen-minute intervals on a low-voltage feeder, 15% of them prosumers, and evaluate the full token life-cycle of minting, settlement, and burning on a single-node validator. The main finding is that the execution layer is not the binding constraint: the community-month replays 1,858× faster than real time with zero delivery loss, and all fifteen energy- and token-conservation assertions close exactly against on-chain state. Participation is instead decided by the price rule, which ranks the uniform-price auction above the regulated buy-back and the CDA.

1. Motivation and Design Rationale

Thailand's state utilities have advanced a National Energy Trading Platform (NETP) since 2017 to facilitate P2P trading among rooftop-solar prosumers, using a Tendermint-based consortium blockchain [1]. This article presents a Solana-based architecture as an alternative execution layer, motivated by NETP's own finding that Tendermint outperformed Ethereum and Hyperledger Fabric in throughput testing, which suggests that further gains may be available from Solana's parallel-execution model. P2P market design and blockchain-based energy trading are already well surveyed [2], [3] and demonstrated in field pilots such as the Brooklyn Microgrid [4]; what such work reports is throughput and market outcome, seldom evidence that the ledger conserves energy and currency exactly. This paper supplies that evidence. The settlement logic is built as six programs on the Anchor framework, Solana's official framework, written in Rust, compiled to SBF bytecode, and executed on the Solana Virtual Machine (SVM). By routing high-frequency writes to per-entity program-derived accounts (PDAs), transactions with disjoint write sets never contend for the same account lock and therefore execute in parallel under Solana's Sealevel runtime. The design claim this paper tests is exactly that: with hot-path state partitioned per entity, what remains of the scaling limit is consensus and lock serialization, not on-chain computation.

2. On-Chain Settlement Layer and Simulated Grid

The evaluated system has two parts, the settlement layer under test and the simulated grid that drives it:

- The smart-contract layer is built on the Solana Anchor framework [5] (anchor-lang and anchor-spl, version 1.0.0) and comprises six programs, **energy-token, governance, oracle, registry, trading,** and **treasury**, all written in Rust and compiled to SBF bytecode. One program invokes another through cross-program invocation (CPI), with authority delegated to program-derived addresses (PDAs) that sign without any private key existing. The deployment targets a permissioned environment realised as a private `solana-test-validator` network; all experiments reported here ran on a single node (Apple M2, Agave 3.1.10).
- The fleet uses the CINELDI 80-bus rural LV reference grid (SINTEF) [6] for network topology only, the 80-node single-phase feeder with its V/A/W parameters, imported from MATPOWER, translated to a GridLAB-D (v5.3.0) model, and resolved for AC power flow with pandapower [7]. As CINELDI carries only hourly load and no generation, all time-series are simulator-produced: per-meter consumption and twelve prosumer nodes' (15%) rooftop PV at fifteen-minute cadence, with solar modelled deterministically via pvlib [8] under a fixed seed.

A complete run must satisfy three closure identities, each checkable from chain state alone:

$$\sum_{i,d} M(i,d) = E_{\text{settle}} = E_{\text{burn}}, \quad \sum_{i} R(i) = \frac{\text{REC supply}}{10^3}, \quad \sum_{m} T_m = P_{\text{sell}} + W \tag{1}$$

Here $M(i,d)$ is the GRID minted for prosumer i on day d; E_{settle} and E_{burn} the energy moved by settlement and retired by burns; $R(i)$ the REC certified to i; and T_m, P_{sell}, W the per-match buyer debit, seller proceeds, and wheeling charges. An audit harness re-derives the fifteen assertions behind Eq. (1) from live chain state by RPC reads alone: per-meter counters, PDA censuses, registry conservation sums, token supplies and balances.

3. Throughput, Audit, and Revenue Results

Throughput is client-observed confirmed goodput, that is confirmed transactions per wall-clock second from burst start to last confirmation. Every non-confirmed transaction is attributed to a failure class, submission reject, on-chain guard, or expiry, separating delivery loss from validation working as intended.

3.1. Simulated Community-Month Dataset

Table 1 summarises the workload every reported figure derives from.

Table 1: The simulated community-month dataset (seed-deterministic; no field data).

Quantity	Value
fleet	80 meters (seeded solar/load physics): 12 prosumer sellers, 68 consumers
market policy	sell cap $C = 10$ kWh/day per prosumer; 0.1 kWh dust floor
horizon / readings	30 days × 96 ticks (Δt = 15 min) = 230,400 readings, integer Wh
month energy	15,868.5 kWh generated; 144,879.8 kWh consumed; 10,386.7 kWh interval surplus
oracle-accepted surplus	8,253.502 kWh; 919 readings gate-rejected

3.2. Measured Outcomes and Limitations

Table 2 reports the replay through the deployed programs. Three findings follow: the execution layer is not the binding constraint, since settlement uses 107 k of the 1.4 M CU per-transaction ceiling and the limit is account-lock serialization and consensus; settlement is exactly auditable, as Eq. (1) holds on chain under all three price rules; and the price rule rather than the platform decides participation, through the wheeling charge that separates those rules. That ranking holds while prosumer supply is capped near demand; left uncapped, P2P net revenue falls below the buy-back, which bears no wheeling charge.

Table 2: Community-month replay outcomes, each figure re-derived from on-chain state.

Metric	Value	What it bounds
replay wall-clock	1,394.7 s for 230,400 readings (1,858× real time)	end-to-end lifecycle cost
telemetry ingest	≈213 readings·s^{-1}	oracle write path under PDA partition
order bursts	≈53 TPS confirmed goodput	trading account-lock serialization
settlement compute	107 k CU per Ed25519-verified match	headroom to the 1.4 M CU ceiling
delivery loss	0 unconfirmed; 919 readings gate-rejected	validation, not lost throughput
audit	15/15 assertions, all three price rules	exactness of settlement
prosumer net revenue	uniform 2.290 > buy-back 2.200 > CDA 2.065 ฿/kWh	P2P participation threshold

Goodput on a single validator bounds execution and account locking only; it is not a consensus-throughput result for a three-utility consortium. The telemetry outcome reproduced bit-identically across four runs, throughput varying 213–232 readings·s^{-1} with host load, while the lifecycle figures come from one canonical run and the price-rule comparison from a separate sweep, since per-order and per-trade nullifier PDAs make an identical replay impossible on a persistent ledger. The dataset is simulator-produced, with no field measurements.

Conclusion

The measurements support the central design claim: with hot-path state in per-entity PDAs, the residual scaling limit is consensus and lock serialization rather than on-chain computation, and settlement is exactly auditable. What remains open is economic, since the wheeling charge, not the ledger, sets the threshold at which a Thai prosumer prefers P2P trading to the buy-back. Extending the evaluation to a multi-validator consortium mapped to EGAT, MEA, and PEA is the natural next step, well suited to UEC–ASEAN collaboration.

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